Data Analytics

Exercise 2: Aggregate Functions & Grouping

employees

id	first_name	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

1. Write a SQL query to find the total number of employees in the company

```
SELECT COUNT(Id) AS total_number_of_employees
From   employees;
```

total_number_of_employees
10

2. Write a SQL query to find the total salary paid to all employees in the IT department.

```
SELECT SUM(Salary) AS total_salary
FROM   Employees
WHERE  department = 'IT';
```

→

total_salary
220000

3. Write a SQL query to calculate the average salary of employees in the HR department

```
SELECT  AVG(Salary) AS Average_salary
From    employees
WHERE   department = 'HR';
```

→

Average_Salary
49 500

4. Write a SQL query to find the highest and lowest salary in the company

```
SELECT MAX(salary) As highest_salary,
       MIN(salary) AS lowest_salary
FROM   employees;
```

highest_salary	lowest_salary
62000	48000

5. Write a SQL query to group employees by department and display the total salary paid in each department.

```
SELECT SUM(salary) AS total_salary,
       department
FROM   employees
Group by department;
```

→

total_salary	department
220000	IT
99000	HR
119000	Finance
105000	Marketing

6. Write a SQL query to count how many employees work in each city.

```
SELECT COUNT(id) AS number_of_employees,
       City
FROM   employees
Group by City;
```

number_of_employees	City
2	New York
3	Chicago
2	Los Angeles
2	San Francisco
1	Houston

7. Write a SQL query to group employees by department, calculate the average salary in each department, and order the result in descending order of average salary

```
SELECT  Avg(salary) AS average_salary,
        department
FROM    employees
Group by department
Order by average_salary desc;
```

average_salary	department
59000	Finance
55000	IT
52500	Marketing
49500	HR

8. Write a SQL query to find departments where the total salary paid exceeds 100000

```
SELECT SUM(salary) AS total_salary,
       department
from   employees
Group by department
Having SUM(salary) > 100000;
```

total_salary	department
220000	IT
119000	Finance
105000	Marketing

9. Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order.

```
SELECT COUNT(id) AS number_of_employees,
       City
from   employees
Group by City
Having COUNT(id) >1
order by number_of_employees desc;
```

number_of_employees	City
3	Chicago
2	New York
2	Los Angeles
2	San Francisco

10. Write a SQL query to find the department with the highest average salary

```
SELECT AVG(salary) AS average_salary,
       department
FROM   employees
Group by department
Order by average_salary desc
Limit 1;
```

→

average_salary	department
59000	Finance